

■ Drone Telemetry Viewer App (Android)

Project Overview

The Drone Telemetry Viewer App is an Android application that connects to a MAVLink telemetry stream (UDP/TCP) and displays live drone telemetry data in real time using modern Android development practices.

Features

- Connect to MAVLink telemetry using UDP or TCP
- Live telemetry dashboard (Latitude, Longitude, Altitude, Battery, Flight Mode)
- Armed / Disarmed status monitoring
- Auto reconnection handling
- Simulated drone action buttons (Arm, Disarm, Takeoff, RTL)
- Robust error handling with crash-free execution

Tech Stack

Language: Kotlin

UI: Jetpack Compose, Material 3

Architecture: MVVM + Clean Architecture

Async: Coroutines, Flow, StateFlow

Dependency Injection: Hilt

Telemetry: MAVLink / MAVSDK

IDE: Android Studio

Architecture Overview

UI (Jetpack Compose) → ViewModel → Use Cases → Repository → MAVLink Data Source → Network Layer (UDP/TCP)

Project Structure

- data – MAVLink client, telemetry parsing, repositories
- domain – models, repositories, use cases
- ui – connection screen and telemetry dashboard
- di – dependency injection modules
- MainActivity – app entry point

Setup & Usage

1. Clone the GitHub repository
2. Open the project in Android Studio
3. Start a MAVLink telemetry source (drone or simulator)
4. Enter IP address and port (example: udp://0.0.0.0:14550)
5. Connect and view live telemetry data

Error Handling

- Invalid IP or port validation
- Connection timeout handling
- Stream interruption recovery
- Null or unsupported MAVLink packets
- Automatic retry and reconnection logic

Future Enhancements

- Google Maps integration with live drone position
- Telemetry data logging using Room database
- Offline telemetry caching
- Unit and UI testing
- Real MAVSDK command execution

Learning Outcomes

- Real-time telemetry streaming in Android
- MAVLink protocol understanding
- Clean Architecture implementation
- Jetpack Compose UI development
- Network resilience and error handling

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